

TLDR Auditory duration judgments are pulled linearly toward visual duration; bias scales with auditory noise but shows no non-linear breakdown even at 250 ms conflicts.

11 participants · 4 tasks · 5 models 2,156 bimodal trials / subj

Question

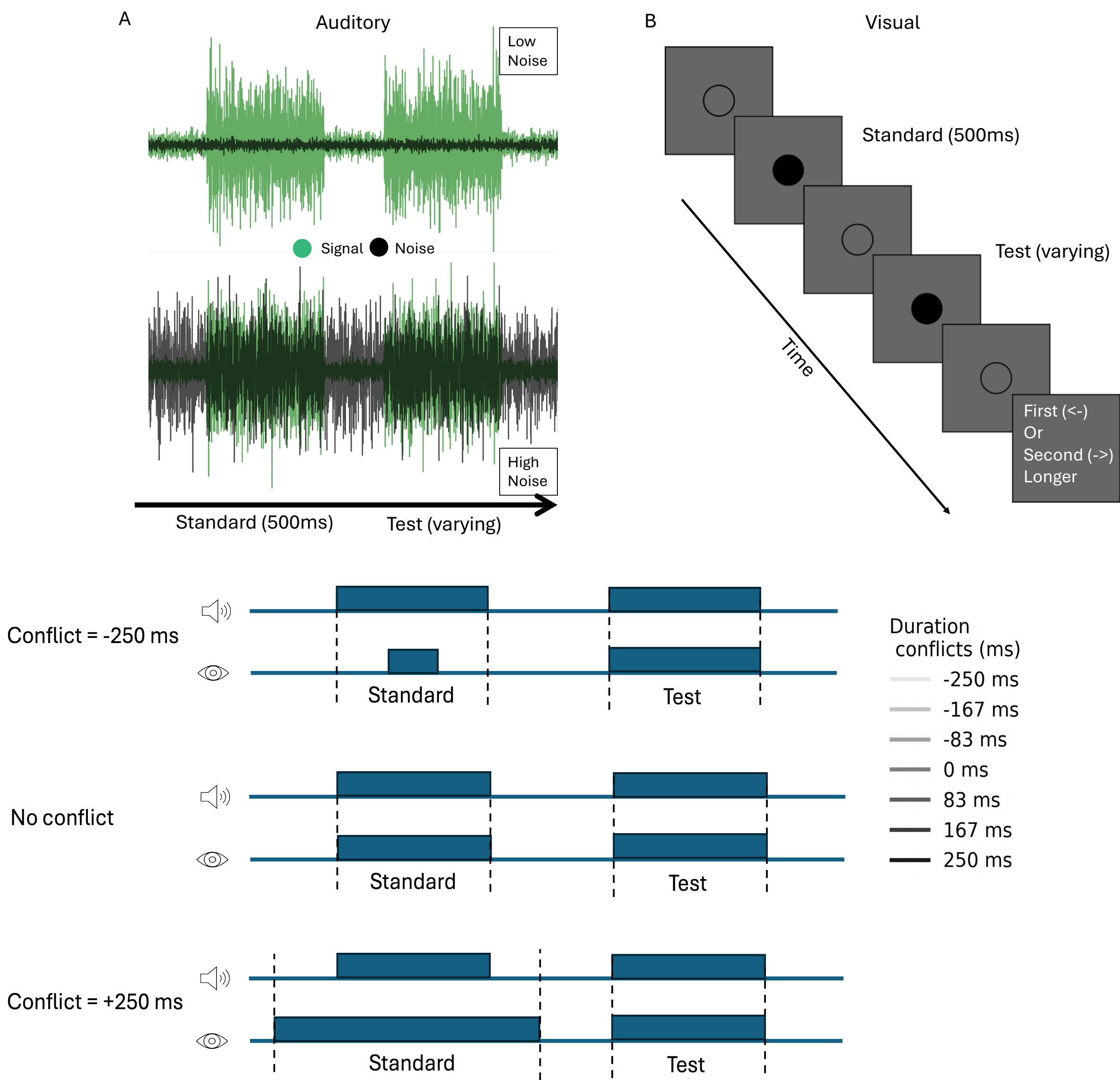
Audiovisual duration cues integrate optimally under small conflicts (Hartcher-O'Brien et al., 2014).

Does causal inference break that integration when AV conflicts grow large, the way it does in spatial localization etc (Kording...).

- Does the brain fuse, segregate, or switch between AV duration cues at large conflict?
- How does auditory reliability shape the visual bias on perceived auditory duration?
- Can behavioral fits actually discriminate these computational strategies?

02 Stimuli & Tasks

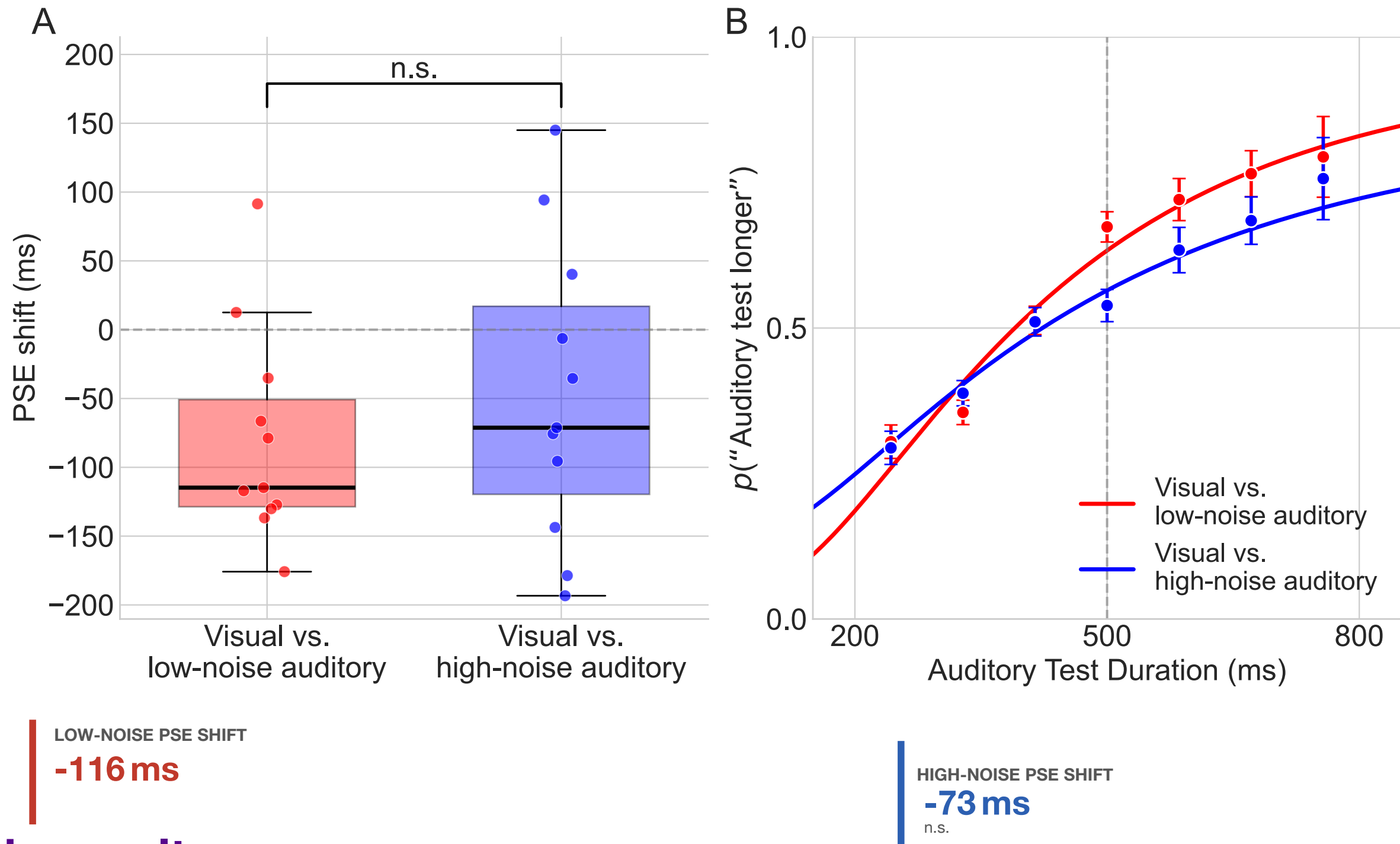
2-IFC · 500 ms standard



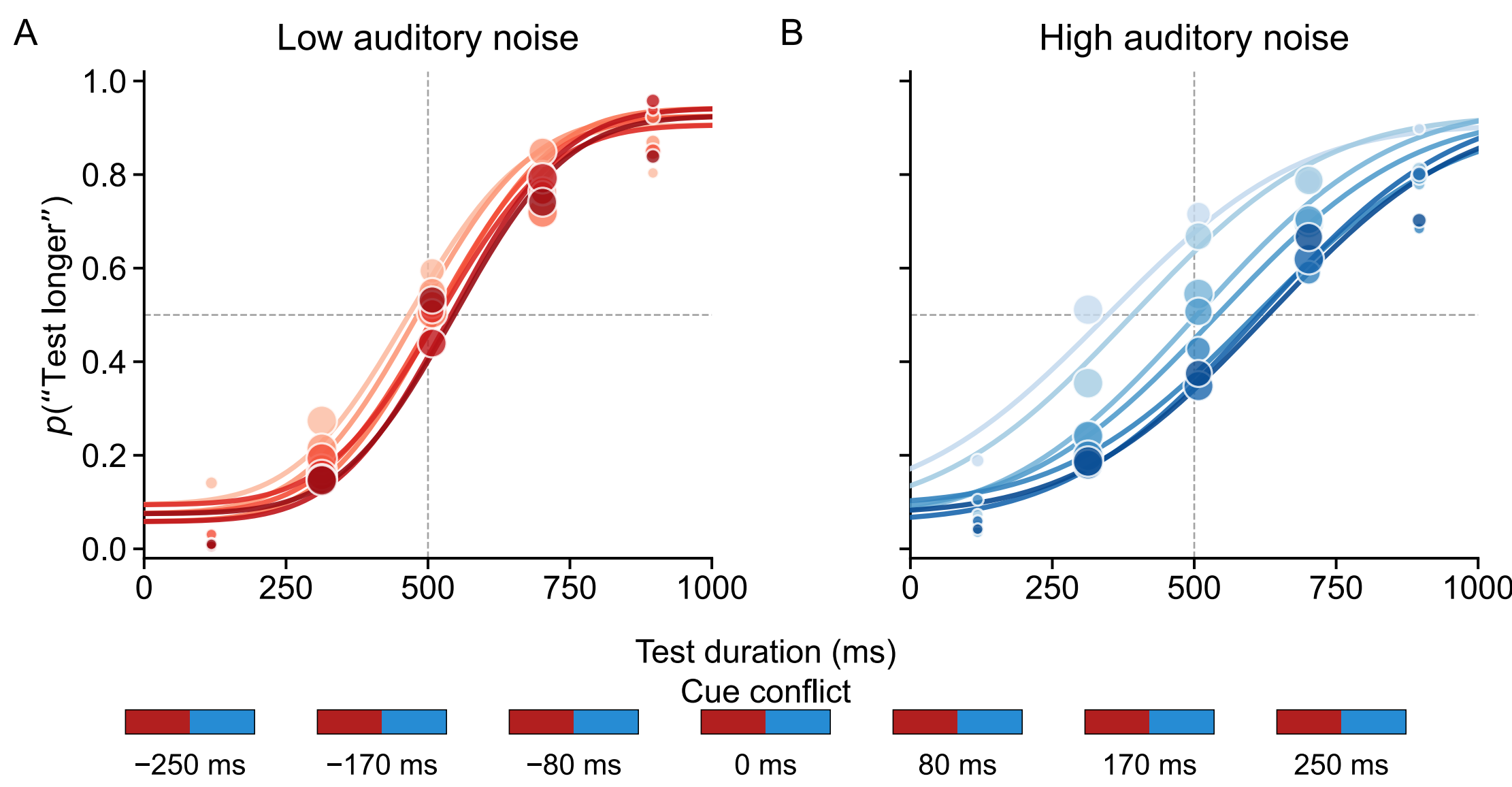
04 Modality bias

cross-modal calibration

Before adding AV conflict we estimate each subject's perceptual auditory-visual bias from a cross-modal task and correct it on the visual stimulus, so the nominal 0-conflict trial is perceptually matched.



05 Main results



05 Models

M1 Forced fusion

M2 CI · Model averaging

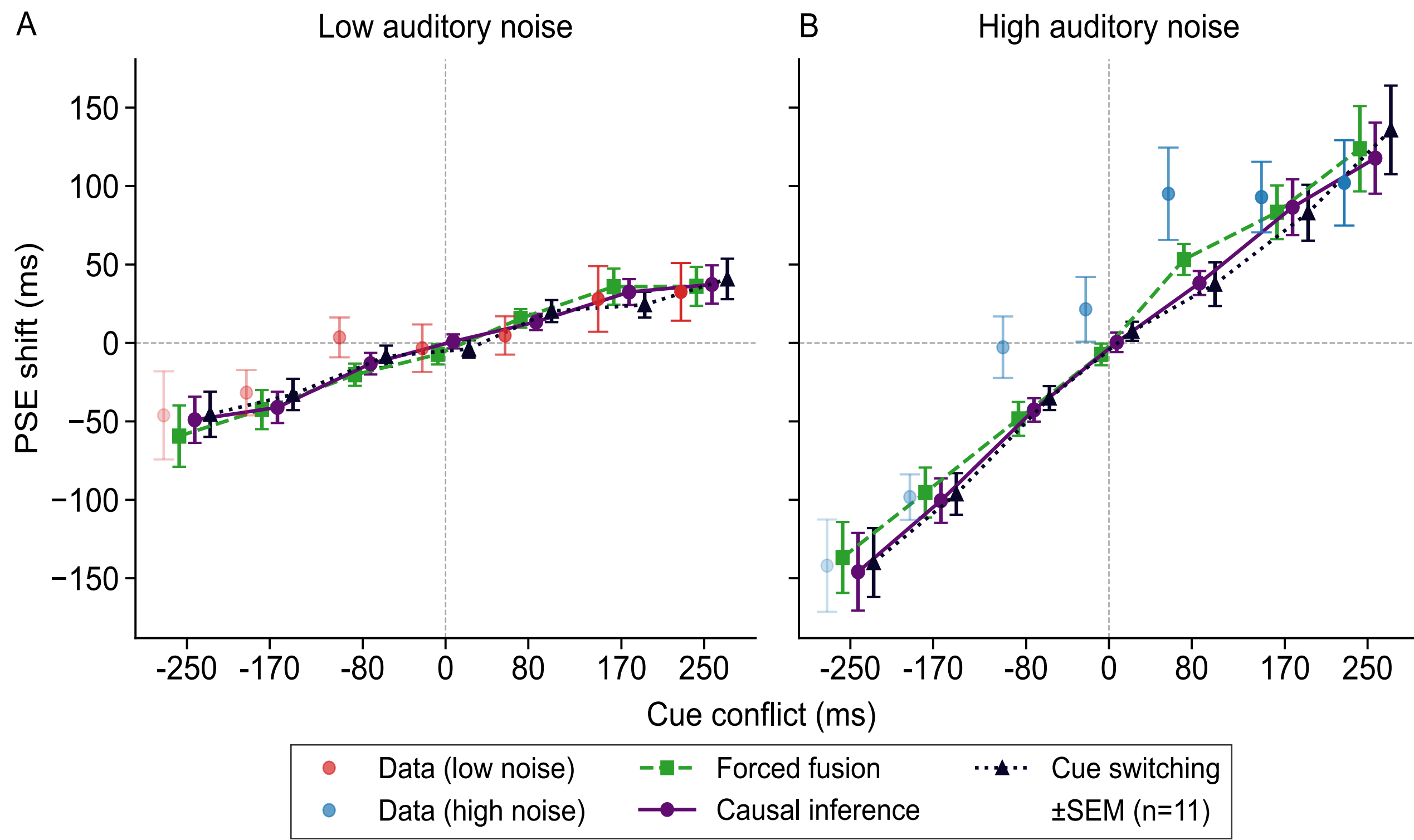
M3 CI · Model selection

M4 CI · Probability matching

M5 Probabilistic cue switching

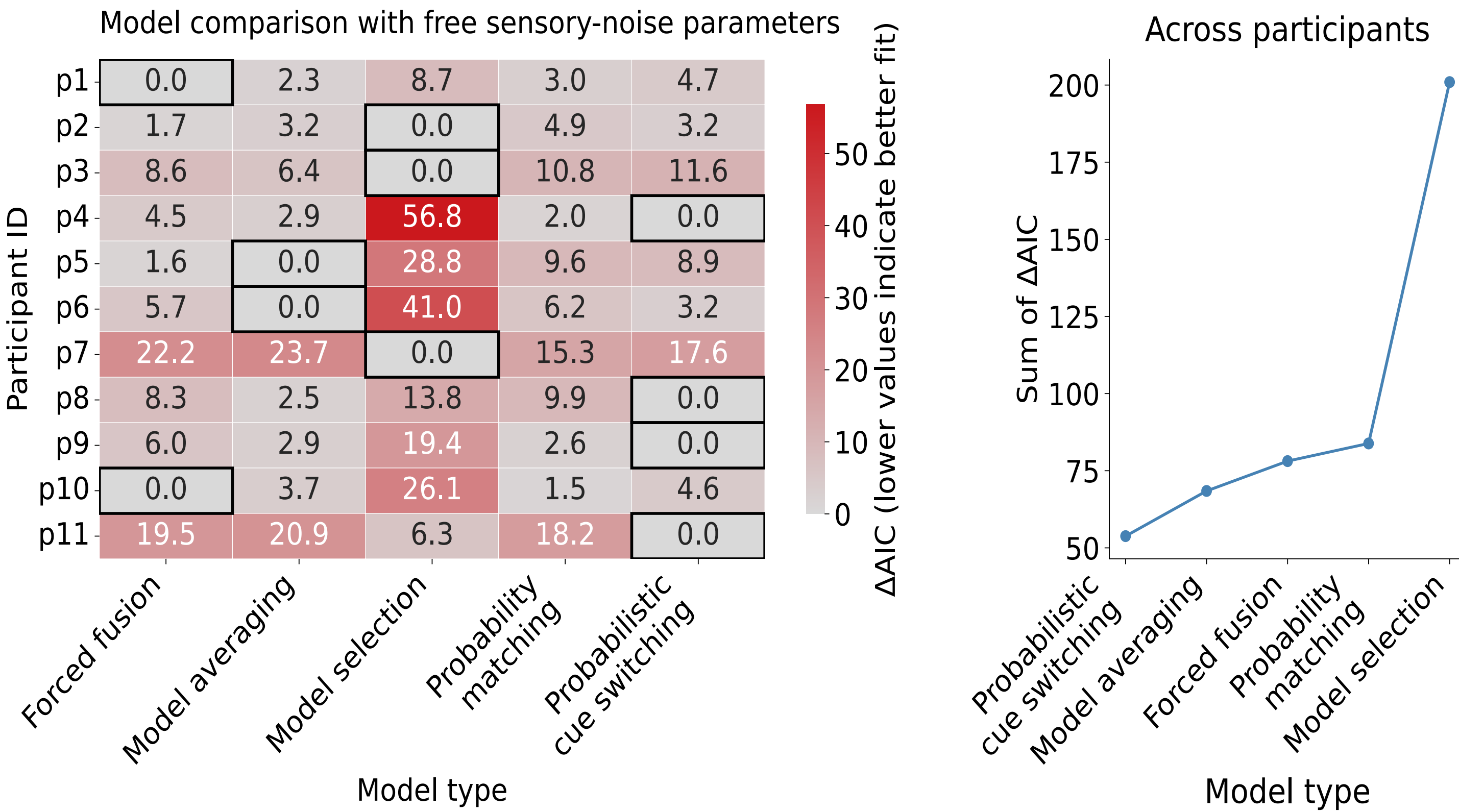
06 Main result · PSE shifts

Visual bias on perceived auditory duration grows approximately linearly with conflict and is much larger under high auditory noise. The non-linear breakdown predicted by causal inference is not observed.



08 Model comparison

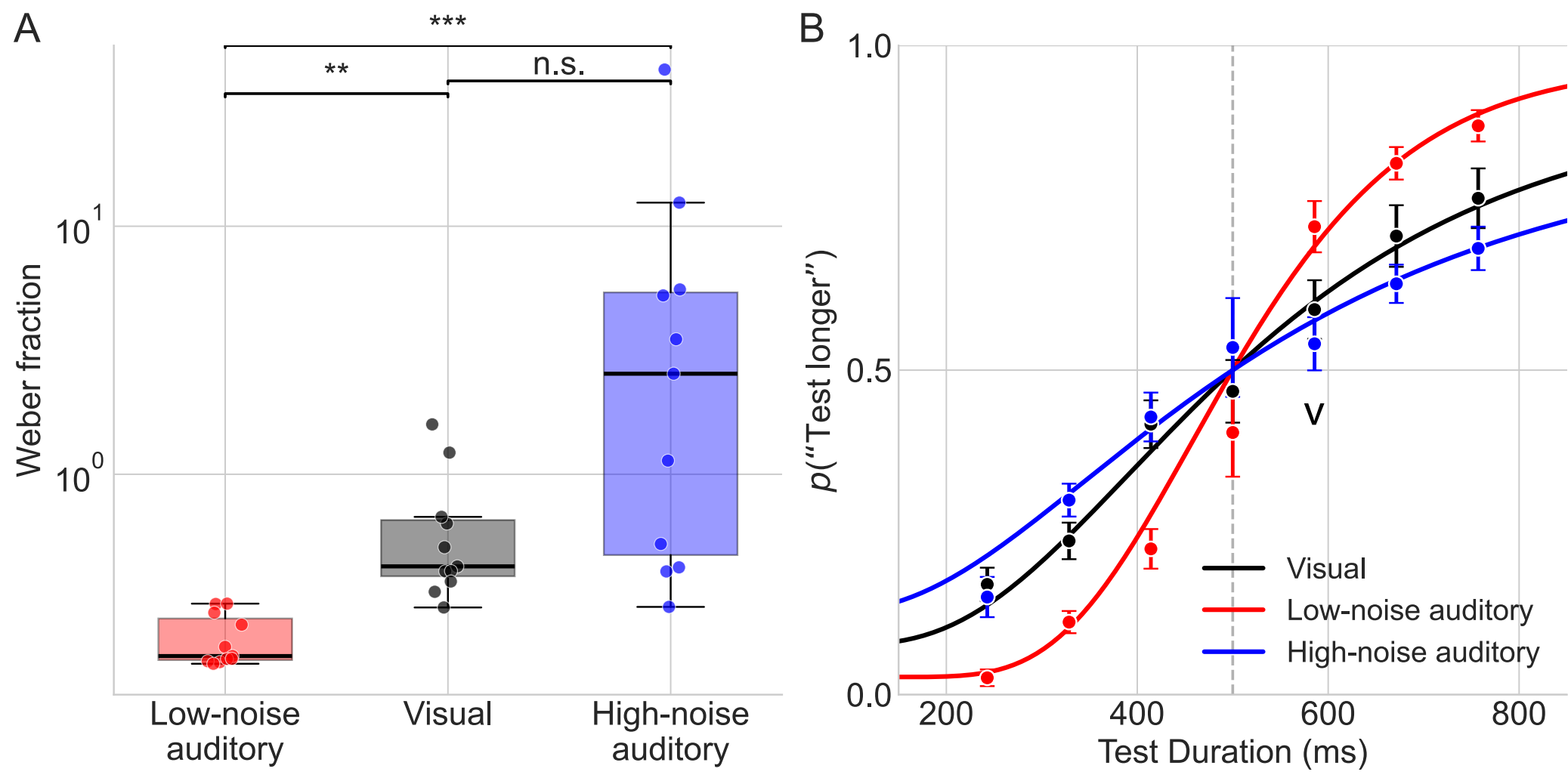
ΔAIC · individual + group



09 Conclusion

- Auditory duration is biased toward vision; bias scales with auditory noise.
- Bias grows linearly with conflict; no causal-inference breakdown observed.
- Heuristic cue switching fits marginally best, but model evidence is essentially flat.
- High encoding noise keeps P(C=1) high, creating an identifiability limit for sub-second AV duration tasks.
- Causal inference may still be plausible; behavior alone cannot adjudicate within experimentally tolerable conflict ranges.

03 Cuereliability



0.18

WF · low-noise aud.

0.42

WF · visual

2.54

WF · high-noise aud.